

YEAR 7 MATHEMATICS

Chapter 10

Analysing data

WORKSHEETS

10

How to use this booklet

- 1** There is **one worksheet for each lesson**. It has three columns that get harder from left to right: **Easy**, then **Medium**, then **Challenging**.

- 2** Read the **speech bubble** at the top of each column: it tells you what to do. A new bubble means a new kind of question.

WE DO **Top of each column.** Your teacher works the examples on the board. **Copy every step** into the space.

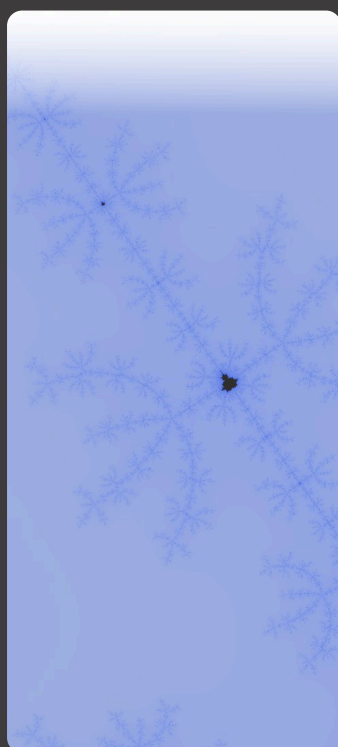
YOU DO **The rest of the column.** Your turn. Column 1: answer on the line. Columns 2 and 3: show your working on the lines.

Contents and progress tracker

Lesson	Topic	Page	Date	Done
10.01	Interpreting graphs	3		☐
10.02	Misleading graphs	4		☐
10.03	Dot plots	5		☐
10.04	Stem-and-leaf plots	6		☐
10.05	Mean and mode	7		☐
10.06	Median and range	8		☐
10.07	Analysing dot plots and stem-and-leaf plots	9		☐
10.08	Comparing data	10		☐

In this chapter you will

- interpret picture, column, sector and line graphs
- identify misleading graphs
- read and draw dot plots and stem-and-leaf plots
- find the mean, mode, median and range
- analyse data shown in plots
- compare two sets of data



About the artwork

The pictures in this booklet are parts of the **Mandelbrot set**, drawn by a computer that repeats one simple rule, $z \rightarrow z^2 + c$, thousands of times for every point. Every booklet shows a different part of the set: this one is snowflake-like branches between the main body and the top bulb. Maths can make beautiful things.

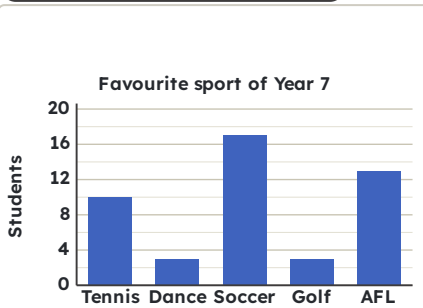
Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

Easy

Questions 1–15

Use the column graph. Read each column against the scale.



WE DO With your teacher: copy the working.

E1 How many chose Tennis? _____

E2 How many chose Dance? _____

YOU DO Your turn.

1 How many chose Soccer? 17

2 How many chose Golf? _____

3 How many chose AFL? _____

4 How many students altogether? _____

5 Which sport was most popular? _____

Reading scales: the column ends halfway between two gridlines. What value does it show?

WE DO With your teacher: copy the working.

E3 between 25 and 30 _____

YOU DO Your turn.

6 between 2 and 4 3

7 between 160 and 180 _____

8 between 10 and 12 _____

9 between 40 and 50 _____

10 between 80 and 90 _____

11 between 100 and 120 _____

12 between 20 and 24 _____

13 between 60 and 80 _____

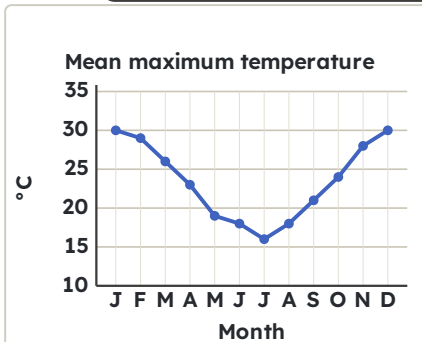
14 between 50 and 100 _____

15 between 6 and 8 _____

Medium

Questions 16–22

Use the line graph. Read the value, then work out the change.



WE DO With your teacher: copy the working.

E1 From January to April _____

E2 From February to September _____

YOU DO Your turn.

16 From March to February

March: 26°C, February: 29°C

change: 3°C up

17 From April to July

April: 23°C, July: 16°C

change: ___°C down

18 From May to December

May: ___°C, December: ___°C

change: ___°C up

19 From June to May _____

20 From July to October _____

21 From August to March _____

22 From September to August _____

Challenging

Questions 23–22

Use the column graph in column 1. Answer each question; give fractions in simplest form.

WE DO With your teacher: copy the working.

E1 How many more chose Tennis than Dance? What percentage of the class is that (1 d.p.)? _____

E2 What fraction, and what percentage (1 d.p.), chose Tennis? _____

YOU DO Your turn.

Easy

Questions 1–12

Misleading (M) or fair (F)?

WE DO With your teacher: copy the working.

- E1** The vertical axis starts at 0. _____
- E2** The vertical axis starts at 90. _____

YOU DO Your turn.

- 1** The columns are different widths. **M** _____

- 2** The scale has equal steps. _____

- 3** The scale jumps 10, 50, 60. _____

- 4** A picture is enlarged in two directions. _____

- 5** It has a title and labelled axes. _____

- 6** It is drawn in 3D so the front looks bigger. _____

- 7** Each picture stands for the same amount. _____

- 8** The axis has no numbers. _____

- 9** The sectors add to 100%. _____

- 10** The sectors add to 130%. _____

- 11** There is a break mark on the axis. _____

- 12** Some months are left out. _____

Medium

Questions 13–16

What makes the graph misleading? Write the problem, then how to fix it.

WE DO With your teacher: copy the working.

- E1** The y-axis goes 0, 5, 10, 50, 100. _____

- E2** The y-axis starts at 1000. _____

YOU DO Your turn.

- 13** A picture twice as tall is also twice as wide.

problem: picture looks 4 times as big
fix: enlarge in one direction only

- 14** The axes have no titles.

problem: missing labels
fix: _____

- 15** Only the months with rising sales are shown.

problem: _____
fix: _____

- 16** A 3D sector graph is tilted. _____

Challenging

Questions 17–21

The axis starts above 0. How many times taller does column B look? How many times bigger is it really (1 d.p.)?

WE DO With your teacher: copy the working.

- E1** A = 92, B = 96, axis starts at 90 _____

- E2** A = 51, B = 54, axis starts at 50 _____

YOU DO Your turn.

- 17** A = 22, B = 30, axis starts at 20

drawn: $10 \div 2 = 5$
real: $___ \div ___ = ___$

- 18** A = 105, B = 110, axis starts at 100

drawn: $___ \div ___ = ___$
real: $___ \div ___ = ___$

- 19** A = 72, B = 80, axis starts at 70 _____

- 20** A = 11, B = 14, axis starts at 10 _____

- 21** A = 62, B = 68, axis starts at 60 _____

Easy

Questions 1–30

Find the mode: the value that occurs most often.



WE DO

With your teacher: copy the working.

E1 7, 8, 6, 4, 8, 8, 3

E2 8, 4, 1, 6, 4, 7, 4

YOU DO

Your turn.

1 2, 7, 3, 2, 4, 2, 2

2

2 5, 4, 6, 8, 1, 7, 8

3 1, 9, 2, 4, 2, 2, 4

4 5, 7, 9, 9, 3, 9, 7

5 7, 8, 9, 7, 7, 9, 1

6 1, 2, 3, 8, 8, 9, 4

7 2, 2, 3, 3, 2, 9, 4

8 3, 6, 3, 3, 2, 8, 3

9 2, 3, 3, 5, 2, 6, 2

10 1, 7, 3, 9, 1, 1, 5

11 7, 6, 6, 4, 2, 3, 5

12 1, 5, 1, 3, 4, 3, 3

13 4, 4, 2, 4, 7, 5, 6

14 4, 4, 7, 1, 1, 8, 4

15 2, 1, 1, 3, 8, 6, 9

16 9, 4, 6, 5, 2, 2, 8

17 3, 8, 5, 1, 6, 3, 3

18 4, 3, 9, 8, 3, 1, 7

19 8, 2, 5, 5, 5, 9, 4

20 2, 1, 9, 7, 5, 8, 1

21 3, 7, 6, 8, 4, 4, 4

22 9, 3, 7, 7, 1, 7, 6

23 7, 4, 9, 4, 9, 9, 1

24 5, 5, 8, 1, 3, 4, 7

25 8, 2, 8, 8, 8, 7, 3

26 4, 6, 4, 6, 6, 3, 6

27 1, 2, 2, 2, 7, 2, 1

28 2, 3, 1, 6, 3, 3, 8

29 2, 6, 3, 5, 9, 9, 1

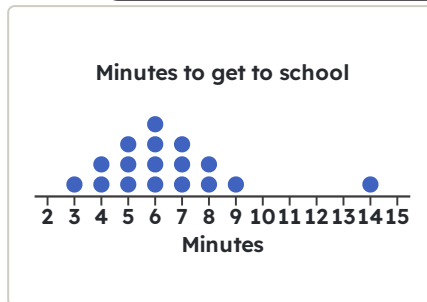
30 6, 4, 4, 1, 3, 7, 5

Medium

Questions 31–34



Use the dot plot. Count the dots carefully.



WE DO

With your teacher: copy the working.

E1 How many values?

E2 What is the mode?

YOU DO

Your turn.

31 What is the range?

14 – 3

= 11

32 What is the smallest value?

count from the plot

=

33 Which value is an outlier?

=

34 How many took more than 6 minutes?

Challenging

Questions 35–41

Draw a dot plot on the back of the sheet, then find the mode and the range.



WE DO

With your teacher: copy the working.

E1 5, 1, 4, 8, 5, 4, 4, 6, 8, 3

E2 1, 3, 8, 4, 6, 5, 8, 5, 5, 1

YOU DO

Your turn.

35 7, 5, 1, 5, 3, 5, 3, 3, 6, 6

ordered: 1, 3, 3, 3, 5, 5, 5, 6, 6, 7

mode: 3 and 5

range: ____ – ____ = ____

36 6, 4, 2, 3, 3, 8, 4, 5, 6, 6

ordered: _____

mode: ____

range: ____ – ____ = ____

37 1, 6, 4, 3, 6, 7, 5, 1, 3, 2

38 8, 3, 3, 2, 6, 6, 6, 3, 8, 3

39 1, 8, 3, 2, 6, 1, 4, 5, 1, 5

40 4, 7, 3, 1, 6, 6, 4, 1, 8, 1

41 5, 7, 2, 6, 8, 6, 6, 2, 7, 2

YEAR 7 • CHAPTER 10: ANALYSING DATA

30	9 3
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34 How many scores are 50 or more?

39 19, 22, 41, 46, 28, 17, 48, 35

Easy

Questions 1–29

Find the mean: add the values, then divide by how many there are.



WE DO

With your teacher: copy the working.

E1 6, 9, 14, 11

E2 12, 5, 20, 7, 11

YOU DO

Your turn.

1 4, 13, 11, 8

9

2 3, 14, 16

3 1, 18, 17

4 17, 16, 4, 11

5 8, 14, 5

6 10, 11, 13, 10

7 15, 16, 13, 4

8 10, 8, 17, 17

9 2, 11, 20

10 16, 19, 16

11 7, 2, 12

12 10, 11, 1, 10

13 4, 9, 5

14 16, 14, 10, 11, 9

15 3, 5, 14, 14

16 19, 6, 8

17 17, 20, 11

18 18, 1, 16, 17

19 16, 2, 18

20 14, 17, 14, 11, 4

21 17, 11, 9, 15

22 19, 1, 11, 17

23 20, 12, 4

24 7, 8, 15

25 11, 4, 9

26 12, 7, 20

27 18, 2, 4

28 9, 18, 3

29 4, 16, 4, 20

Medium

Questions 30–39

Find the mean. Show the total and the division. Round to 1 d.p.



WE DO

With your teacher: copy the working.

E1 25, 39, 39, 22

E2 19, 20, 40, 30, 24, 29

YOU DO

Your turn.

30 9, 27, 33, 35, 31, 23

total: $9 + 27 + 33 + 35 + 31 + 23 = 158$

mean: $158 \div 6 = 26.3$

31 8, 24, 20, 21, 25, 10

total: $8 + 24 + 20 + 21 + 25 + 10 = 108$

mean: $___ \div ___ = ___$

32 35, 37, 12, 8

total: $___ + ___ + ___ + ___ = ___$

mean: $___ \div ___ = ___$

33 24, 9, 6, 19, 21, 22

34 35, 9, 33, 23, 18

35 8, 20, 29, 11

36 11, 37, 23, 36, 35

37 11, 7, 31, 14, 39

38 10, 14, 26, 9, 23, 26

39 19, 26, 32, 17, 37, 11

Challenging

Questions 40–46

The mean is given. Find the missing value. Work out the total first.



WE DO

With your teacher: copy the working.

E1 5, 1, 1, ? (mean 7)

E2 13, 22, 1, 31, ? (mean 17)

YOU DO

Your turn.

40 14, 19, ? (mean 16)

total must be $16 \times 3 = 48$

known: 33

missing: $___ - ___ = ___$

41 7, 3, 10, ? (mean 9)

total must be $___ \times ___ = ___$

known: $___$

missing: $___ - ___ = ___$

42 12, 4, ? (mean 9)

43 2, 8, ? (mean 7)

44 8, 22, 22, 12, ? (mean 16)

45 21, 4, 6, 8, ? (mean 11)

46 6, 9, 5, 18, ? (mean 9)

Easy

Questions 1–25

Find the median: order the values, then take the middle one.

WE DO With your teacher: copy the working.

E1 16, 21, 4, 18, 4

E2 13, 11, 22, 14, 13

YOU DO Your turn.

1 4, 22, 25, 2, 11

11

2 30, 16, 16, 22, 1

3 7, 29, 15, 1, 5

4 18, 8, 3, 23, 20

5 7, 13, 20, 12, 30

6 26, 14, 6, 9, 8

7 17, 22, 12, 1, 10

8 5, 23, 4, 18, 30

9 10, 29, 8, 1, 16

10 7, 14, 27, 10, 26

11 13, 8, 17, 14, 10

12 12, 12, 10, 6, 6

Find the range: largest – smallest.

WE DO With your teacher: copy the working.

E3 8, 38, 38, 28, 14

YOU DO Your turn.

13 10, 24, 46, 16, 16

36

14 36, 27, 27, 33, 43

15 13, 2, 15, 47, 6

16 5, 12, 27, 23, 2

17 48, 32, 49, 22, 32

18 15, 14, 15, 43, 1

19 41, 1, 4, 37, 19

20 25, 22, 3, 27, 18

21 41, 41, 23, 6, 42

22 29, 38, 3, 23, 40

23 30, 19, 3, 26, 41

24 31, 28, 6, 19, 24

25 14, 50, 49, 37, 21

Medium

Questions 26–35

Even number of values: order them, then halve the sum of the middle two.

WE DO With your teacher: copy the working.

E1 24, 27, 13, 36, 31, 26

E2 15, 18, 14, 8, 21, 14

YOU DO Your turn.

26 38, 19, 24, 32, 4, 20

ordered: 4, 19, 20, 24, 32, 38

$(20 + 24) \div 2 = 22$

27 35, 14, 30, 7, 15, 9

ordered: 7, 9, 14, 15, 30, 35

$(\text{ } + \text{ }) \div 2 = \text{ }$

28 13, 37, 23, 18

ordered: $\text{ } , \text{ } , \text{ } , \text{ } , \text{ } , \text{ }$

$(\text{ } + \text{ }) \div 2 = \text{ }$

29 5, 40, 10, 12, 29, 9

30 10, 21, 19, 17

31 28, 11, 31, 9, 22, 26

32 1, 24, 36, 34

33 4, 9, 31, 9

34 33, 37, 25, 8, 31, 15

35 40, 38, 40, 33, 34, 21

Challenging

Questions 36–42

Find the median and the range. Show the ordered list.

WE DO With your teacher: copy the working.

E1 28, 22, 39, 24, 31, 25, 26, 36

E2 60, 57, 30, 54, 39, 24, 24

YOU DO Your turn.

36 45, 54, 59, 33, 44, 48, 19

ordered: 19, 33, 44, 45, 48, 54, 59

median: 45

range: $\text{ } - \text{ } = \text{ }$

37 31, 24, 51, 32, 29, 37, 30

ordered: $\text{ } , \text{ } , \text{ } , \text{ } , \text{ } , \text{ } , \text{ }$

median:

range: $\text{ } - \text{ } = \text{ }$

38 42, 58, 27, 38, 46, 22, 45, 10

39 29, 35, 36, 48, 43, 53

40 10, 55, 18, 42, 59, 35, 54

41 58, 27, 31, 47, 15, 19, 11, 25

42 26, 32, 44, 27, 45, 29, 29

Easy

Questions 1–9

Use the stem-and-leaf plot.

Heights of seedlings (mm)

Stem	Leaf
3	6 6 7
4	2 4
5	0 2 9
6	3
7	1 3

Key: 4 | 5 = 45 mm

WE DO With your teacher: copy the working.

E1 How many values? _____

E2 Lowest value _____

YOU DO Your turn.

1 Highest value 73

2 Range _____

3 Median _____

4 Values below 50 _____

Find the range of each stem-and-leaf plot (key 2 | 3 = 23).

WE DO With your teacher: copy the working.

3 0 5	
4 8	
5 7	
6 2 7	

YOU DO Your turn.

1 | 1 2

2 |

5 3 | 8 9 52

4 | 7

5 |

6 | 3

3 | 1 8 9

6 4 | 7 _____

5 | 1 2

1 | 4 5

2 |

7 3 | _____

4 | 4 4

5 | 8

6 | 4

2 | 9

8 3 | 2 4 6 _____

4 | 9

5 | 6

1 | 8 8

2 |

9 3 | _____

4 | 4 9

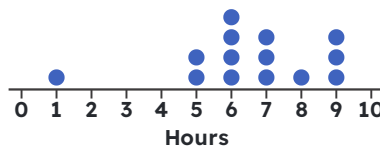
5 | 1 4

Medium

Questions 10–12

Use the dot plot. Show how you found each answer.

Hours of sleep



WE DO With your teacher: copy the working.

E1 Find the median. _____

E2 Find the mean (1 d.p.). _____

YOU DO Your turn.

10 Find the range. _____

9 – 1

= 8

Find the median of the stem-and-leaf plot. Count to the middle leaf.

WE DO With your teacher: copy the working.

E3 1 9
2 6 6 6
3 8
4 5
5 5

YOU DO Your turn.

11 1 3 4 6
2 4 8
3
4
5 3 7

7 values: the 4th

median = 24

12 2 7
3 0 0 9 9
4 2 6

7 values: the 4th

median = ____

Challenging

Questions 13–17

Find the mean and the median with and without the outlier. Which changes most?

WE DO With your teacher: copy the working.

E1 14, 11, 10, 11, 15, 19, 44

E2 11, 11, 15, 13, 12, 13, 53

YOU DO Your turn.

13 11, 13, 15, 13, 11, 11, 50

with: mean 17.7, median 13

without 50: mean 12.3, median 12

14 18, 18, 14, 16, 13, 10, 43

with: mean ____, median ____

without ____: mean ____, median ____

15 16, 13, 18, 18, 15, 17, 55

16 15, 14, 10, 17, 18, 20, 52

17 12, 14, 15, 16, 11, 17, 47

Easy

Questions 1–18

Which group is more consistent? (Smaller range.)

WE DO With your teacher: copy the working.

E1 A: range 18, B: range 25

E2 A: range 20, B: range 28

YOU DO Your turn.

1 A: range 20, B: range 27 **A**

2 A: range 10, B: range 8

3 A: range 18, B: range 21

4 A: range 6, B: range 5

5 A: range 21, B: range 10

6 A: range 28, B: range 22

7 A: range 23, B: range 12

8 A: range 3, B: range 27

9 A: range 12, B: range 27

10 A: range 29, B: range 25

11 A: range 20, B: range 15

12 A: range 15, B: range 22

13 A: range 19, B: range 3

14 A: range 10, B: range 25

15 A: range 16, B: range 3

16 A: range 12, B: range 10

17 A: range 5, B: range 22

18 A: range 29, B: range 19

Medium

Questions 19–24

Find the mean of each group. Which group has the higher mean?

WE DO With your teacher: copy the working.

E1 A: 13, 10, 10, 14
B: 4, 7, 2, 18

E2 A: 4, 16, 16, 18
B: 8, 7, 9, 13

YOU DO Your turn.

19 A: 2, 7, 10, 16
B: 5, 4, 11, 14

A: $35 \div 4 = 8.75$

B: $34 \div 4 = 8.5$

higher mean: A

20 A: 12, 13, 16, 8
B: 10, 11, 3, 17

A: $49 \div 4 = 12.25$

B: $41 \div 4 = 10.25$

higher mean: _____

21 A: 20, 15, 15, 18
B: 4, 18, 14, 4

A: $___ \div ___ = ___$

B: $___ \div ___ = ___$

higher mean: _____

22 A: 6, 3, 2, 18
B: 18, 8, 5, 19

23 A: 16, 7, 18, 16
B: 17, 11, 14, 10

24 A: 15, 7, 10, 5
B: 2, 11, 18, 18

Challenging

Questions 25–29

Compare the two classes: find the mean and the range of each, then write a sentence.

WE DO With your teacher: copy the working.

E1 Class A: 7, 13, 15, 9, 7
Class B: 20, 14, 11, 17, 11

E2 Class A: 12, 19, 20, 13, 9
Class B: 20, 8, 7, 6, 15

YOU DO Your turn.

25 Class A: 7, 7, 13, 14, 11
Class B: 6, 9, 20, 5, 17

A: mean 10.4, range 7

B: mean 11.4, range 15

26 Class A: 13, 7, 10, 6, 13
Class B: 8, 7, 13, 17, 13

A: mean _____, range _____

B: mean _____, range _____

27 Class A: 20, 12, 16, 10, 9
Class B: 9, 6, 18, 11, 12

28 Class A: 14, 8, 7, 5, 20
Class B: 7, 11, 7, 8, 5

29 Class A: 8, 15, 10, 19, 12
Class B: 12, 6, 12, 12, 14